

Project Structure

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01_Project Brief

Radiation Equipment Consultancy Website

Project Overview

Develop a professional, modern, responsive consultancy website for an independent consultant specializing in:

- Radiation Equipment Installation
- Commissioning & Acceptance Testing
- Staff Training
- Radiation Safety
- Quality Assurance
- Project Management
- Regulatory Compliance

The website should present the consultancy professionally to hospitals, cancer centres, universities, government agencies and private healthcare institutions.

The website is NOT an ecommerce platform.

The website is NOT a hospital management system.

The primary objective is to create an online presence that builds trust, showcases expertise and allows potential clients to contact the consultant.

Goals

The website should

- Build credibility
- Present services professionally
- Showcase equipment expertise
- Improve Google visibility
- Allow clients to request consultations
- Allow all website content to be managed from an Admin Dashboard

Technology Stack

Frontend

HTML5

Bootstrap 5

CSS3

JavaScript

FontAwesome

AOS Animation Library

Backend

PHP 8.x

Database

MySQL

Hosting

Hostinger

Web Server

Apache

General Requirements

Responsive Design

SEO Friendly

Fast Loading

Modern UI

CMS Driven

Secure

Easy to Maintain

Professional Medical Design

02 Solution Design Document (SDD)

Solution Design Document

Architecture

The application consists of two independent components.

1. Public Website

Accessible by everyone.

Displays all information.

2. Admin Dashboard

Accessible only by authenticated administrators.

Used to manage all website content.

Both use the same MySQL database.

Architecture

Browser



PHP Application



MySQL Database



Admin Dashboard

Functional Modules

Public Website

[Home](#)

[About](#)

[Services](#)

[Equipment](#)

[Blog](#)

[Contact](#)

[Privacy](#)

[Terms](#)

[404](#)

Admin Dashboard

[Dashboard](#)

[Homepage Manager](#)

[About Manager](#)

[Service Manager](#)

[Equipment Manager](#)

[Blog Manager](#)

[Testimonials Manager](#)

[Media Library](#)

Contact Messages

SEO Settings

Website Settings

Admin Profile

Logout

Content Management Philosophy

The layout of the website is fixed.

The content is dynamic.

Every heading

Every paragraph

Every image

Every button

Every service

Every testimonial

Every SEO setting

must come from the database.

No business content should be hardcoded into HTML.

Security

Prepared Statements

Password Hashing

Session Authentication

Role Based Access

CSRF Protection

Input Validation

Image Validation

SQL Injection Protection

XSS Protection

Performance

Bootstrap Minified

Lazy Loaded Images

Optimized CSS

Optimized JavaScript

Responsive Images

Caching Headers

Compressed Assets

User Types

Administrator

Can edit everything.

No public registration.

No client accounts.

Website Sections

Homepage

Hero

About Preview

Services

Equipment

Why Choose Us

Process

CTA

Footer

Each section editable.

Contact Workflow

Visitor submits contact form.

↓

Stored in database.



Email notification sent.



Visible inside dashboard.



Administrator replies externally.

Blog Workflow

Administrator creates article.



Publishes.



Visible immediately.



SEO metadata stored.

Image Workflow

Administrator uploads image.



Stored inside uploads/



Database stores filename.



Frontend loads dynamically.

03 Technical Design Document (TDD)

Technical Design Document

Folder Structure

/

admin/

assets/

css/

js/

images/

uploads/

includes/

config/

database/

controllers/

models/

views/

api/

index.php

Database Tables

admin_users

website_settings

homepage

about

services

equipment

blog_categories

blog_posts

testimonials

messages

seo

media

activity_logs

Coding Standards

PHP 8

Object Oriented

PDO Only

Prepared Statements

Reusable Functions

Reusable Components

No duplicated code

MVC-inspired organization

Includes

header.php

navbar.php

footer.php

sidebar.php

db.php

auth.php

functions.php

Frontend Components

Navbar

Hero

Cards

Timeline

Accordion

Contact Form

Footer

Blog Cards

Buttons

Breadcrumbs

Modals

Backend Components

Authentication

CRUD Engine

Image Upload

Email Engine

SEO Manager

Media Manager

Dashboard Widgets

Image Upload Rules

jpg

jpeg

png

webp

Maximum 5MB

Unique filename

Stored inside uploads/

Database Access

PDO only.

Prepared statements only.

No mysqli.

Responsive Breakpoints

Desktop

1200+

Laptop

992

Tablet

768

Mobile

576

Naming Convention

snake_case for database

camelCase for PHP methods

kebab-case for CSS

PascalCase for PHP classes

Admin Theme

Bootstrap 5

Dark Sidebar

Light Content Area

Responsive

Dashboard Cards

Charts

Tables

Modals

Toast Notifications

Public Theme

White

Blue

Medical Style

Large Typography

Rounded Cards

Soft Shadows

Smooth Animations

Minimalistic

Professional

04 MASTER PROMPT FOR CURSOR

You are a Senior Full Stack PHP Software Engineer and UI/UX Designer.

Your task is to build a complete professional consultancy website for an independent Radiation Equipment Consultant.

The project MUST follow the attached Project Brief, Solution Design Document and Technical Design Document exactly.

General Requirements

- Use PHP 8

- Use MySQL
- Use Bootstrap 5
- Use HTML5
- Use CSS3
- Use Vanilla JavaScript
- Use PDO
- Use prepared statements
- Build reusable components
- Use include files
- Never duplicate code
- Build clean maintainable architecture

Website Requirements

The public website must contain

- Home
- About
- Services
- Equipment
- Blog
- Contact

The website must be fully responsive.

Every page must be SEO friendly.

Every image must be optimized.

All animations should be subtle and professional.

Admin Dashboard

Build a secure admin dashboard that manages ALL website content.

The administrator must be able to edit

Homepage

About

Services

Equipment

Blog

Testimonials

Contact Information

SEO

Website Settings

without touching code.

Do NOT hardcode any business content inside HTML.

The frontend should retrieve all content dynamically from MySQL.

The only hardcoded elements should be the layout and reusable components.

The admin dashboard should provide complete CRUD functionality.

Use Bootstrap tables.

Use Bootstrap forms.

Use Bootstrap cards.

Use Bootstrap modals.

Use Bootstrap alerts.

Project Quality

Write production-quality code.

Comment complex code.

Use clean folder organization.

Avoid unnecessary JavaScript frameworks.

Avoid unnecessary PHP frameworks.

The final application should be deployable directly to Hostinger with minimal configuration.

Implementation Strategy

Implement the project module by module.

Start by creating:

1. Project folder structure
2. Database schema
3. Configuration files
4. Authentication system
5. Admin dashboard layout

6. Public website layout

Only after the foundation is complete should you implement the CMS modules.

Never skip steps.

After completing each module, ensure the application remains fully functional before proceeding to the next one.